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Project Final Report

Camera Lidar Debris Analyzer (CLiDA)

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Contents

1	Project Purpose	6
2	Project Objectives and Functional Requirements	6
2.1	Specific Objectives	7
2.2	Concept of Operations	7
2.3	Functional Block Diagram	8
2.4	Functional Requirements	9
3	Final Design	9
3.1	Design Requirements Flow-Down	9
3.2	Final Design Overview	11
3.3	Subsystems and Components	12
3.3.1	LiDAR Sensor	12
3.3.2	Camera	12
3.3.3	Mounting and Stabilization System	13
3.3.4	Processing Unit	14
3.4	System Operation	14
3.5	Software Architecture	16
3.6	Summary of Final Design	18
4	Manufacturing	18
4.1	Scope of Manufacturing Tasks	18
4.1.1	Hardware Components	18
4.1.2	Software Development	18
4.2	Outcome of Manufacturing Tasks	19
4.3	System Integration into a Functioning Unit	19
5	Verification and Validation	19
5.1	Test Stand Verification	20
5.1.1	Availability of Precise 3D Dimensions	20
5.1.2	Controlled Data Acquisition Timing	20
5.1.3	Fixed Distance Maintenance	20
5.1.4	High-Fidelity Data Storage	21
5.1.5	Rotation Capabilities and Data Visualization	21
5.1.6	Rotation About Multiple Principal Axes	21
5.2	Software Validation	22
5.2.1	Data Pre-processing and Automated Execution	22
5.2.2	Zero-Rotation and Rotation Identification	22
5.2.3	Shape and Dimension Recognition	23
5.2.4	Rotational Speed Estimation	23
5.3	Troubleshooting and Mitigation Strategies	23
5.3.1	Sensor Calibration and Alignment	24
5.3.2	Environmental Interference	24
5.3.3	Data Processing Efficiency	24
5.3.4	Incomplete Sensor Integration Due to Equipment Delays	24

5.4	Validation Against Functional Requirements	24
5.5	Verification and Validation Results	25
6	Risk Assessment and Mitigation	25
6.1	Realized Risks and Their Impact	25
6.1.1	Errors in Software Interpretation of Simulated Rotational Data	25
6.2	Lessons Learned from Risk Management	25
6.3	Alternative Actions	25
7	Project Planning	26
7.1	Organizational Structure	26
7.2	Work Breakdown Structure (WBS)	26
7.3	Work Plan (WP)	27
7.4	Cost Plan (CP)	28
7.5	Test Plan (TP)	29
7.5.1	Specialized Equipment and Facilities	29
7.5.2	Data Collection and Analysis Procedures	30
7.5.3	Communication and Documentation	30
7.6	Risk Management Integration	30
7.7	Project Planning Summary	30
8	Lessons Learned	31
8.1	Importance of Early Integration	31
8.2	Effective Communication and Team Coordination	31
8.3	Flexibility and Adaptability	31
8.4	Thorough Testing and Validation	32
8.5	Challenges of Sensor Integration	32
8.6	Realistic Simulation Environments	32
8.7	Time Management and Scheduling	32
8.8	Advice for Future Teams	33
8.9	Summary of Lessons Learned	33
9	Individual Report Contributions	33
9.1	Jerry Jack	33
9.2	Janga Bliss	33
9.3	Elijah Smith	34
9.4	Sahra Genc	34
9.5	Jack Grimes	34
10	Appendices	35
10.1	Key Design Alternatives	35
10.1.1	Design Option 1: Flash lidar	35
10.1.2	Design Option 2: Camera with simple lidar	35
10.2	Trade Study Process and Results	36

List of Figures

1	Concept of operations	7
2	Functional Block Diagram	8
3	Flash LiDAR's components disassembled and cad view.	12
4	Actual image of the camera.	13
5	CLiDA Mounting Setup: LiDAR mounted on optical table with camera attached via custom 3D printed plate	14
6	Experimental Setup	16
7	Dimensions of Triangle and Rectangle	20
8	Using Caliper for Truth Values	20
9	Comparison of the object's appearance in the camera image (left) versus its representation in the flash LiDAR point cloud (right), illustrating how each sensor modality captures distinct information about the object's spatial presence.	21
10	Rotation of the object about various principal axes, namely flat, upright, and standard, in that order.	22
11	Comparison of raw and processed point cloud data. Post-processing eliminates noise and outliers, leaving a clean and focused point cloud of the rectangular object. . . .	22
12	Example of software output indicating whether the dataset represents a non-rotating object ("is not moving") or a rotating object ("is moving") based on changes in consecutive frames.	23
13	Work Breakdown Structure for the CLiDA Project	27
14	Gantt Chart for Project Timeline.	28
15	Estimated Project Budget and Expenses	28

List of Tables

1	Specific objectives table	7
2	Functional Requirements	9
3	Key specifications of the flash LiDAR.	12
4	Key specifications of the camera.	13
5	Rectangle dimension measurement error and likelihood	23
6	Measured Rotation Rates vs. Actual Values	23
7	Team Roles and Responsibilities	26
8	Comprehensive Test Plan for CLiDA System Validation	29
9	Pros and cons tables for design option 1	35
10	Pros and cons tables for design option 2	35
11	Consideration and Weights for Design Options	36
12	Metric Ranking	37
13	Trade Study Evaluation	38

List of Acronyms

- **ADRV**: Active Debris Removal Vehicle
- **CARS**: Capture and Release System
- **CLiDA**: Camera Lidar Debris Analyzer
- **DOCS**: Debris Object Characterization System
- **DR**: Design Requirement
- **ESA**: European Space Agency
- **FR**: Functional Requirement
- **ICP**: Iterative Closest Point
- **LiDAR**: Light Detection and Ranging
- **NASA**: National Aeronautics and Space Administration
- **PCA**: Principal Component Analysis

Definition of Symbols (Nomenclature)

θ	Rotation angle (degrees)
ω	Rotational speed (degrees per second)
x, y, z	Spatial coordinates in 3D space
$\mathbf{R}$	Rotation matrix
$\mathbf{T}$	Transformation matrix
λ	Wavelength of the LiDAR laser (nm)
v	Velocity (m/s)

1 Project Purpose

The rapid increase of space debris presents a growing hazard to operational satellites, space stations, and future space missions[1]. Since the launch of Sputnik in 1957, more than 4,000 rocket launches and 150 in-orbit fragmentation events have contributed to the space-debris environment [2]. Today, over 8,700 objects larger than 10 cm are in orbit, and approximately 94% of these objects are non-functional, categorized as space debris[3] [4]. Among this debris are smaller components that are challenging to detect and analyze due to their complex shapes, motion, and conditions [5] [6].

Efforts to mitigate space debris have gained increasing attention. NASA’s Active Debris Removal Vehicle (ADRV) uses a debris object characterization system (DOCS) to calculate debris speed, trajectory, and rotation rates, combined with a capture and release system (CARS) to remove non-cooperative targets [7]. The European Space Agency’s partnership with Clear Space focuses on a four-arm claw mechanism to latch onto debris and deorbit it safely [8]. Airbus has developed a net system capable of collecting objects up to two meters in diameter for controlled atmospheric burn-up [9].

The Camera Lidar Debris Analyzer (CLiDA) aims to explore similar challenges, focusing on objects smaller than 15 cm. By using a lidar system, CLiDA will work to identify and characterize smaller debris components. The project will focus on determining basic information such as the size, rotation axis, and speed of these objects to support tracking and potential future removal operations.

By developing CLiDA, our team hopes to contribute to understanding small debris in space and support safer, more effective debris management efforts. This system will serve as a stepping stone for further exploration and refinement in addressing the growing challenge of space debris.

2 Project Objectives and Functional Requirements

In this section, we outline the specific objectives, functional requirements, and operational framework for the Camera Lidar Debris Analyzer (CLiDA). Building upon the project purpose discussed earlier, we detail the measurable goals and technical expectations that guide the development of CLiDA. This includes functionality, software algorithms, power systems, and camera capabilities. The concept of operations and functional block diagram further clarify how these elements interact to achieve the project’s mission. By defining these parameters, we establish a clear path toward creating a system capable of identifying and analyzing space debris effectively.

2.1 Specific Objectives

Table 1: Specific objectives table

Functionality	
Level 1	The system shall be able to detect an object with lidar and camera.
Level 2	The system shall be able identify the shape and the size of the object.
Level 3	The system shall be able to identify the rotational axis and the rotational speed of the object.
Software/ Algorithm	
Level 1	The algorithm shall be able to interpret the LiDAR data and output numerical values of the range, size, and the shape of an object.
Level 2	The algorithm shall be able to interpret the lidar data and output a 2D height map of the object.
Level 3	The algorithm shall be able to interpret the lidar data and output a 3D model of the object.
Camera	
Level 1	The camera shall be able to capture images of the object in monochrome with 720p resolution.
Level 2	The camera shall be able to capture images of the object in RGB with 2mp resolution.
Level 3	The camera shall be able to capture images of the object with auto zoom and 10mp resolution.

2.2 Concept of Operations

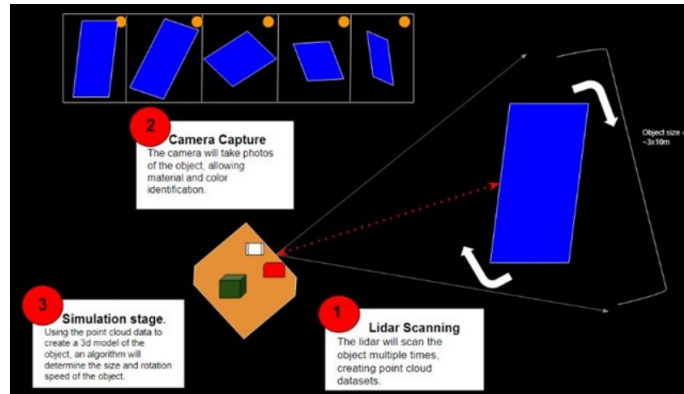


Figure 1: Concept of operations

The Concept of Operations (ConOps) (Figure 1) for the Camera Lidar Debris Analyzer (CLiDA) illustrates how the system operates to analyze space debris. The process begins with Lidar Scanning, where the onboard LiDAR system captures detailed point cloud data by scanning the debris multiple times. This data provides crucial information about the debris' range, size, and shape. Next, during the Camera Capture phase, high-resolution images are taken to identify the debris material, color, and surface details. Finally, in the Simulation Stage, the gathered LiDAR and camera data are combined to generate a comprehensive 3D model of the object. This model is used to calculate the object's size, rotation axis, and speed. The ConOps highlights CLiDA's role in providing accurate debris characterization for future removal operations.

2.3 Functional Block Diagram

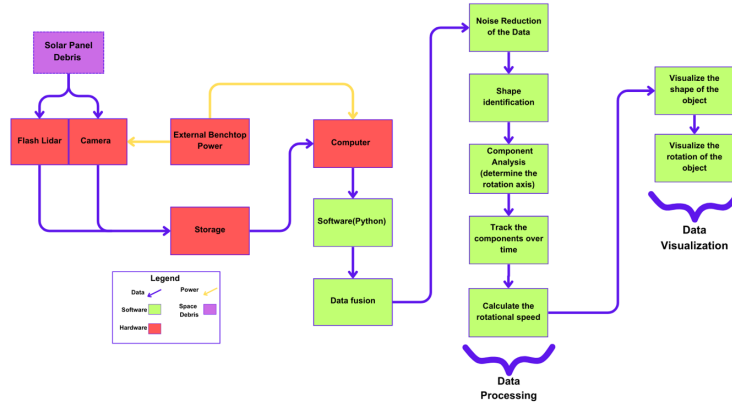


Figure 2: Functional Block Diagram

The Functional Block Diagram (Figure 2) illustrates the key components and workflow within the CLiDA system. Starting from the left, solar panel debris is captured by the flash lidar and camera, powered by an external benchtop source. Data from these sensors is stored and then processed on a computer running Python software for data fusion. The data processing stage involves noise reduction, shape identification, component analysis to find the rotation axis, tracking of component movements, and calculation of rotational speed. Finally, the processed data is used for visualizing the shape and rotation of the object, providing insights for debris analysis.

2.4 Functional Requirements

Table 2: Functional Requirements

FR	Name	Description
FR1	Structural Integrity	CLiDA shall be securely mounted and stable for the duration of its operation.
FR2	Operation	CLiDA shall be capable of collecting and storing data of the object regardless of background.
FR3	Shape Recognition	CLiDA shall be capable of identifying the dimensions of the object.
FR4	Simulation	CLiDA shall be capable of modeling the rotation of the object in 3D.
FR5	Interface	CLiDA shall provide a user-friendly interface for straightforward operation of the software.

This section provides a comprehensive understanding of CLiDA’s objectives and functional requirements, forming the basis for its technical design and implementation. The detailed objectives and diagrams ensure clarity in execution and alignment with the overall project goals. In the next section, we delve into the system’s final design, translating these requirements into tangible engineering solutions.

3 Final Design

In this section, we present the final design of CLiDA, detailing how the system fulfills the functional requirements through specific design solutions. We describe the requirements flow-down from high-level functional requirements to detailed design requirements, providing the rationale for each major requirement. Additionally, we provide a comprehensive overview of the system’s design, including key parameters, subsystems, and operational principles. Engineering design details are illustrated through diagrams and images to convey a thorough understanding of all critical design elements and corresponding solutions.

3.1 Design Requirements Flow-Down

The design of CLiDA is driven by a set of functional requirements (FRs), each of which is broken down into detailed design requirements (DRs). This flow-down ensures that each aspect of the system is purposefully designed to meet the project’s objectives. Below, we outline the flow-down from functional requirements to design requirements, explaining the rationale behind each.

Functional Requirement 1 (FR1): Structural Integrity

FR1: *CLiDA shall be securely mounted and stable for the duration of its operation.*

- **Rationale:** Stability is crucial to minimize vibrations and external disturbances that could affect the LiDAR readings. Accurate data collection depends on the system maintaining a consistent position relative to the target object.
- **Design Requirements Derived from FR1:**

- **DR1.1:** CLiDA shall include a mounting system designed to withstand operational stresses such as vibration and environmental factors.
 - * *Rationale:* A robust mounting system prevents displacement or vibrations that could introduce errors in measurements.
- **DR1.2:** CLiDA’s mounting system shall ensure continuous operation without manual adjustments for stability.
 - * *Rationale:* Eliminating the need for manual adjustments reduces the risk of human error and ensures uninterrupted data collection.
- **DR1.3:** CLiDA shall maintain stability throughout the data collection.
 - * *Rationale:* Consistent stability over time is necessary to guarantee reliable and repeatable results.

Functional Requirement 2 (FR2): Data Collection and Storage

FR2: *CLiDA shall be capable of collecting and storing data of the object regardless of background.*

- **Rationale:** The system must reliably detect and analyze objects in various environments, ensuring that background interference does not compromise data quality.
- **Design Requirements Derived from FR2:**
 - **DR2.1:** CLiDA shall distinguish the object from background noise.
 - * *Rationale:* Effective discrimination between the target object and background is essential for accurate data analysis.
 - **DR2.2:** CLiDA shall have sensitivity adjustments to optimize object detection in different environments.
 - * *Rationale:* Adjusting sensitivity allows the system to perform effectively under varying environmental conditions, such as different lighting or backgrounds.
 - **DR2.3:** CLiDA shall collect data under a variety of environmental conditions and object orientations.
 - * *Rationale:* The system’s versatility in different scenarios enhances its applicability and robustness.

Functional Requirement 3 (FR3): Shape Recognition

FR3: *CLiDA shall be capable of identifying the dimensions of the object.*

- **Rationale:** Accurately determining the object’s dimensions is critical for calculating its rotation and for any potential future removal operations.
- **Design Requirements Derived from FR3:**
 - **DR3.1:** CLiDA shall determine the object’s outline, shape, and color.
 - * *Rationale:* Determining the object’s outline, shape, and color provides comprehensive data to identify the object and understand its physical characteristics for further processing.

Functional Requirement 4 (FR4): 3D Rotation Modeling

FR4: *CLiDA shall be capable of modeling the rotation of the object in 3D.*

- **Rationale:** Understanding the object’s rotational dynamics is essential for tracking and potential future removal operations.
- **Design Requirements Derived from FR4:**
 - **DR4.1:** CLiDA shall construct a 3D model of the object from LiDAR and camera data.
 - * *Rationale:* A detailed 3D model allows for precise analysis of the object’s shape and aids in rotation calculation.
 - **DR4.2:** CLiDA shall record the object’s rotation over time.
 - * *Rationale:* Capturing temporal changes in rotation provides data on angular velocity and rotational axis, critical for dynamic analysis.

Functional Requirement 5 (FR5): User Interface

FR5: *CLiDA shall provide a user-friendly interface for straightforward operation of the software.*

- **Rationale:** A user-friendly interface enhances usability, reduces the likelihood of operator error, and facilitates efficient operation.
- **Design Requirements Derived from FR5:**
 - **DR5.1:** The CLiDA interface shall provide access to functions with streamlined steps.
 - * *Rationale:* Simplified navigation ensures that users can operate the system effectively without extensive training.

3.2 Final Design Overview

The CLiDA system integrates hardware and software components to detect, analyze, and model space debris objects, particularly focusing on objects smaller than 15 cm, such as solar panel fragments. The system comprises the following major subsystems:

- **LiDAR Sensor:** A flash LiDAR unit for capturing 3D point cloud data of the target object.
- **Camera:** A camera for capturing visual images of the object, providing color and texture information.
- **Mounting and Stabilization System:** A mount to securely hold the LiDAR and camera, ensuring system stability.
- **Processing Unit:** A computer system running algorithms, processing sensor data to generate 3D models and calculate rotational dynamics.
- **User Interface:** Software interface allowing users to control the system and visualize results.

3.3 Subsystems and Components

3.3.1 LiDAR Sensor

Description: The flash LiDAR sensor captures 3D point cloud data by emitting laser pulses and measuring the time-of-flight of reflected signals. We selected this flash LiDAR because it aligns with our budget, offers advanced data visualization through its dedicated software, and supports preprocessing capabilities. This allows us to efficiently filter and manage point cloud data, enhancing the accuracy and usability of the captured information for our application.

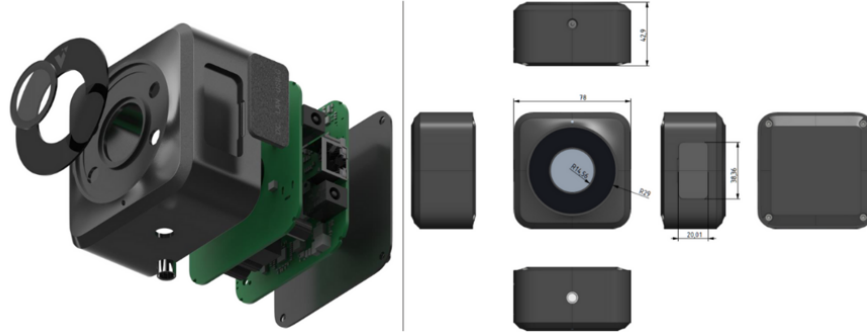


Figure 3: Flash LiDAR's components disassembled and cad view.

Table 3: Key specifications of the flash LiDAR.

Specifications	
Range	Up to 10 meters
Field of View	0.22 degrees horizontal, 0.24 degrees vertical
Resolution	128 × 64 pixels per frame
Frame Rate	1-20 FPS
Voltage	12-25V (DC) @ 3A

3.3.2 Camera

Description: A RGB camera captures images of the target object, providing color, brightness, and texture information. This specific camera was chosen for its auto-focus capability and USB interface, which enables seamless integration and simultaneous processing of LiDAR and camera data, enhancing the efficiency and precision of our system.



Figure 4: Actual image of the camera.

Table 4: Key specifications of the camera.

Specifications	
Resolution	1456 × 1088 pixels per frame
Features	Auto-focus, adjustable exposure
Interface	USB 3.0 connection to processing unit
Frame Rate	149 FPS @ 300 MB/s (Mono8)

3.3.3 Mounting and Stabilization System

Description: The CLiDA system’s sensors are securely mounted to ensure stability and proper alignment during operation. The flash LiDAR sensor is directly mounted onto an optical table using screws, providing a stable and vibration-free platform due to the table’s solid construction. The camera is attached to the side of the LiDAR sensor using a custom-designed 3D printed plate, which ensures precise alignment between the two sensors and maintains their relative positions.

Features:

- **Direct Mounting on Optical Table:** The optical table offers a rigid and stable base, minimizing vibrations and external disturbances.
- **Custom 3D Printed Camera Mount:** The camera is secured to the LiDAR sensor with a custom 3D printed plate, designed specifically to fit the dimensions and mounting points of both devices.
- **Simplified Setup:** The straightforward mounting approach simplifies assembly and reduces the potential for misalignment.

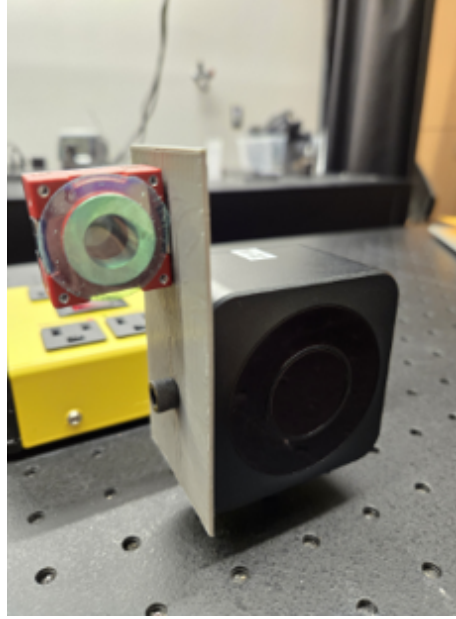


Figure 5: CLiDA Mounting Setup: LiDAR mounted on optical table with camera attached via custom 3D printed plate

3.3.4 Processing Unit

Description: The CLiDA system utilizes a dedicated computer running data processing algorithms developed in Python. It integrates libraries such as Open3D for efficient point cloud processing and ensures streamlined analysis.

Features:

- **Data Acquisition Modules:** Modules are designed for LiDAR and camera data integration to ensure synchronized data collection.
- **Noise Reduction and Filtering:** Implements advanced routines to minimize noise and filter out irrelevant data points.
- **Shape Recognition and Modeling:** Algorithms enable precise recognition of object shapes and creation of 3D models.
- **Rotation Analysis:** Modules determine rotational axes and speeds to enhance object characterization.

3.4 System Operation

The CLiDA system operates through the following main steps:

1. **Data Acquisition:**

- The flash LiDAR sensor collects point cloud data of the target object.
- The LiDAR's built-in software filters the data to isolate the target object from the background.
- Collected data is stored directly onto a flash drive connected to the LiDAR system.

2. Data Retrieval:

- The processing unit accesses the stored data by reading from the flash drive.
- Point cloud data files are organized and prepared for processing.

3. Data Processing and Analysis:

- Preprocessing steps clean and downsample the point cloud data.
 - Data Cleanup: Any data points returned with invalid range values (e.g., NaNs or infinite distances) are filtered out.
 - * Initial points: 76,800
 - * After removing invalid points: 21,784 (approximately 71.6% reduction)
 - Downsampling: The remaining points are then further reduced to create a lower-density point cloud that still represents the object's essential features.
 - * After downsampling: 7,561 points (additional 65.3% reduction from the cleaned set)
- Algorithms calculate the object's dimensions and determine its orientation.
- If sequential data frames are available, the system analyzes changes over time to calculate rotational axis and speed.

4. Output and Visualization:

- Results, including dimensions, orientation, and rotational data, are outputted for further analysis.
- Visualization tools can be used to render the 3D model and display the object's movement.

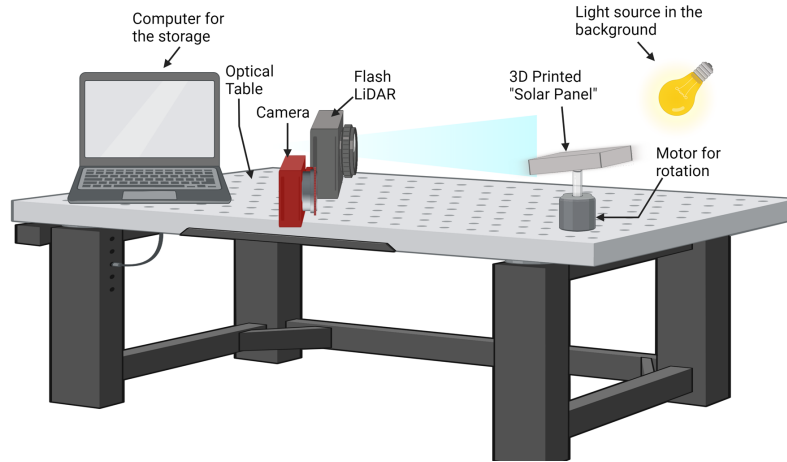


Figure 6: Experimental Setup

3.5 Software Architecture

Overview:

The software component of the CLiDA system is developed in Python and is structured into several modules, each responsible for specific aspects of data processing and analysis. The code utilizes libraries such as Open3D for point cloud processing, NumPy for numerical computations, and the built-in `logging` module for debugging and information output.

- **Preprocessing Module:**

- Loads point cloud data files (.pcd) from the specified directory.
- Removes invalid values such as NaNs and infinite values from the point clouds.
- Applies statistical outlier removal to clean the data.
- Downsamples the point clouds using an estimated voxel size to reduce computational load.
- Estimates surface normals (i.e., vectors perpendicular to local surface patches) for each point to assist in understanding object orientation and shape.

- **Dimension Calculation Module:**

- Calculates the dimensions (width, height, depth) of the object by finding the minimum and maximum bounds along each axis of the point cloud.

- **Orientation Determination Module:**

- Determines the orientation of the object using Principal Component Analysis (PCA) on the point cloud data.
- Identifies the principal axes to understand the object's main orientation in space.

- **Motion Detection Module:**

- Analyzes a sequence of point clouds to determine if the object is moving.
- Computes mean distances between consecutive frames; if distances exceed a threshold, motion is detected.

- **Speed and Rotation Axis Calculation Module:**

- Applies the Iterative Closest Point (ICP) algorithm between consecutive point clouds to estimate transformations.
- Calculates the cumulative transformation to determine the overall rotation of the object.
- Extracts the rotation matrix to compute the rotational speed and rotation axis.
- Provides the rotational speed in degrees per second and estimates the rotation axis direction.

Error Handling and Validation:

- The software includes error handling mechanisms to manage cases where point clouds are empty or invalid.
- Logging statements provide information on the success or failure of each processing step, aiding in debugging and ensuring data integrity.

Visualization and Output:

- The system outputs calculated dimensions, orientation vectors, motion status, rotational speed, and rotation axis.
- Results are printed to the console for immediate feedback.
- Visualization of point clouds and results can be performed using Open3D's visualization tools (though not explicitly shown in the main code).

Alignment with Functional Requirements:

- **FR2 - Data Collection and Storage:**

- The software effectively collects and processes data regardless of background by removing noise and invalid points.

- **FR3 - Shape Recognition:**

- Dimension calculation and orientation determination modules allow the system to identify the object's dimensions and orientation.

- **FR4 - 3D Rotation Modeling:**

- The motion detection and speed calculation modules enable the system to model the object's rotation in 3D.

3.6 Summary of Final Design

The final design of the CLiDA system successfully addresses the project’s objectives and functional requirements. Through careful integration of hardware and software components, the system provides accurate detection, modeling, and analysis of small space debris objects. The design solutions implemented ensure robustness, reliability, and usability, contributing valuable capabilities to support safer and more effective space debris management efforts.

4 Manufacturing

4.1 Scope of Manufacturing Tasks

The manufacturing phase of the CLiDA project involved the integration of purchased components, self-manufactured parts, and software development, covering mechanical and software elements.

4.1.1 Hardware Components

Acquired Components We acquired several essential mechanical components from Fort Lewis College:

- **Servo Motor:** Used to simulate the rotational dynamics of space debris.
- **Optics Table:** Provided a stable, vibration-free platform necessary for precise sensor operation and data accuracy.
- **Screws and Fasteners:** Used for assembling various mechanical parts securely.

Manufactured Components Custom components were manufactured to meet specific project requirements:

- **3D Printed Silhouette:** A silhouette of a small object was 3D printed to act as a controlled test object for the detection and analysis algorithms.
- **3D Printed Mount:** A custom mount was designed and 3D printed to securely hold the lidar sensor, ensuring proper alignment and positioning relative to the test object.

Purchased Components

- **Lidar Sensor:** Purchased online, the lidar provided high-precision point cloud data essential for our analysis.
- **Camera:** Purchased online, the camera was compact, low cost, and possesses a lens mount that minimizes inconsistency and variation.

4.1.2 Software Development

For the software component, we utilized Open3D, an open-source library for 3D data processing. This library facilitated the development of algorithms for point cloud analysis, shape recognition, and visualization.

4.2 Outcome of Manufacturing Tasks

The manufacturing tasks culminated in a fully integrated system capable of meeting the project’s objectives. Key outcomes and challenges included:

- **Hardware Integration:** Successfully combined the servo motor, optics table, and 3D printed components. Precise alignment was achieved to ensure accurate data capture from the lidar sensor.
- **Simulation of Space Debris:** The 3D printed silhouette allowed for realistic simulation of space debris, enabling effective testing of detection and characterization algorithms.
- **Data Accuracy:** Challenges such as calibrating the lidar sensor and minimizing vibrations were addressed through iterative adjustments and testing, enhancing overall system precision.
- **Software Implementation:** Open3D libraries were effectively employed to process and visualize point cloud data. Initial challenges with data noise and processing speed were mitigated by optimizing algorithms and implementing efficient filtering techniques.

These challenges were overcome through collaborative efforts and iterative refinement, resulting in a system that effectively demonstrates the project’s capabilities.

4.3 System Integration into a Functioning Unit

The components were integrated into a functioning system through the following steps:

1. **Hardware Assembly:** The 3D printed mount was used to securely attach the silhouette to the servo motor. The servo motor was set up to rotate the 3D printed silhouette, simulating the motion of space debris.
2. **Electrical Integration:** Connected the lidar sensor and servo motor to the control systems and power supplies, ensuring reliable operation and synchronization between components.
3. **Software Configuration:** Developed software using Open3D libraries to process and visualize data from the lidar sensor. Implemented algorithms for point cloud analysis.
4. **Testing and Calibration:** Conducted extensive testing to calibrate the lidar sensor and validate the performance of the detection and analysis algorithms, ensuring accuracy and reliability.

Through careful manufacturing and integration efforts, the CLiDA system was assembled into a cohesive unit. The combination of mechanical components, precise manufacturing, and robust software development enabled the successful detection, analysis, and characterization of simulated space debris, fulfilling the project’s functional objectives.

5 Verification and Validation

This section details the verification and validation processes undertaken to ensure that the CLiDA system meets its functional requirements and performs reliably. The approach involved a series of carefully designed tests to assess both the hardware (test stand) and software components of the system. The results demonstrate the system’s capability to detect, characterize, and model space debris analogs under controlled conditions.

5.1 Test Stand Verification

5.1.1 Availability of Precise 3D Dimensions

Accurate 3D models of the test object was essential for establishing a ground truth for shape recognition and validation. The rotator object, the rectangular shape, was precisely fabricated using 3D printing technology. High-precision calipers were used to measure the dimensions of the object, ensuring accuracy within ± 0.1 mm. These dimensions served as the baseline for comparing the reconstructed models generated by the system.

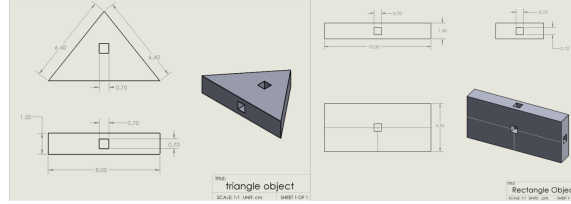


Figure 7: Dimensions of Triangle and Rectangle

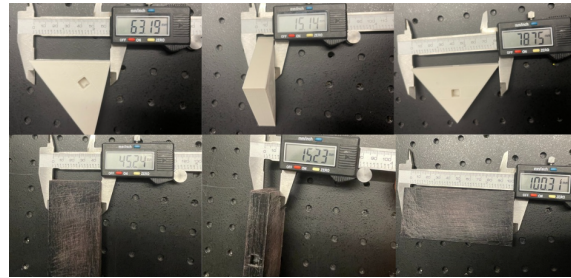


Figure 8: Using Caliper for Truth Values

5.1.2 Controlled Data Acquisition Timing

To enhance the temporal accuracy of data collection, the LiDAR sensor was configured to capture measurements at specific time intervals synchronized with the rotation of the objects. A time-stamping mechanism was implemented within the data acquisition software, allowing for precise control over measurement timing. This facilitated the chronological ordering of data and enabled the pruning of measurements based on time tags during post-processing.

5.1.3 Fixed Distance Maintenance

Maintaining a consistent distance between the rotator and the LiDAR/camera setup was crucial for accurate data capture. The center of mass of each test object was positioned at a fixed distance of 48 cm from the sensors. A meter stick was used for measuring distances, and a motor was screwed into the optics table to hold it in place.

[Video Demonstration of Fixed Distance Maintenance](#)

5.1.4 High-Fidelity Data Storage

All measurements from the LiDAR sensor and camera were stored in their raw formats to preserve data integrity for post-processing. LiDAR data was saved in the standard Point Cloud Data (PCD) format, retaining full spatial resolution and point accuracy. Data backups on Github were made after each testing session to prevent any potential loss.

5.1.5 Rotation Capabilities and Data Visualization

The test stand was designed to rotate the objects around a principal axis with precise control over the rotational speed. Using a calibrated step motor, the objects were rotated at speeds of 1 RPM and 5 RPM. The rotational motion was monitored using a high-resolution rotary encoder with a sampling rate of 10 fps. This setup provided accurate measurements of the phase angle and spin rate, with a standard deviation of less than 0.5 degrees for the angle and 0.05 RPM for the rotation rate over at least 10 full revolutions.

Data visualization was achieved through the Open3D library, which enabled the rendering of point cloud data. Visual representations of the captured data facilitated the verification of the system’s tracking and modeling capabilities.

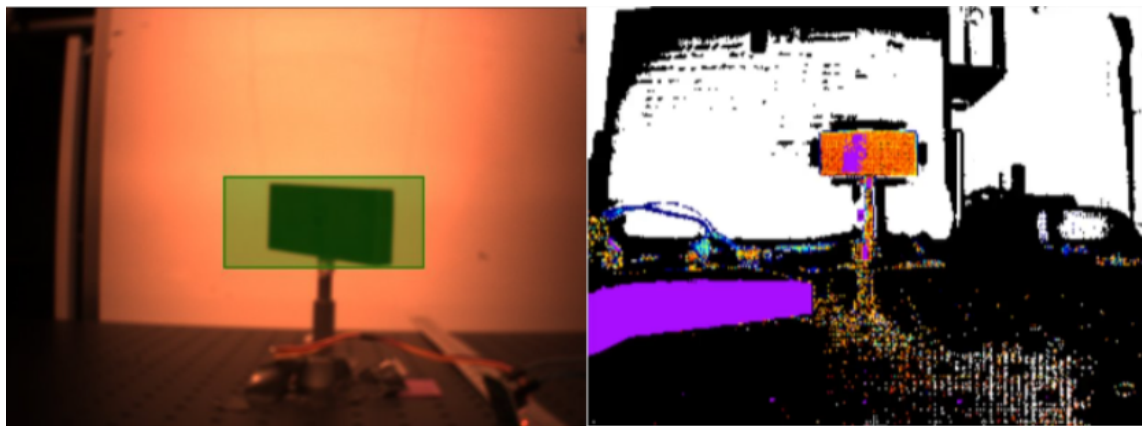


Figure 9: Comparison of the object’s appearance in the camera image (left) versus its representation in the flash LiDAR point cloud (right), illustrating how each sensor modality captures distinct information about the object’s spatial presence.

5.1.6 Rotation About Multiple Principal Axes

The test stand was configured to rotate objects about three different principal axes (one axis at a time). The system successfully captured and processed data for each type of axis rotation, demonstrating its ability to handle multiple distinct rotational orientations.

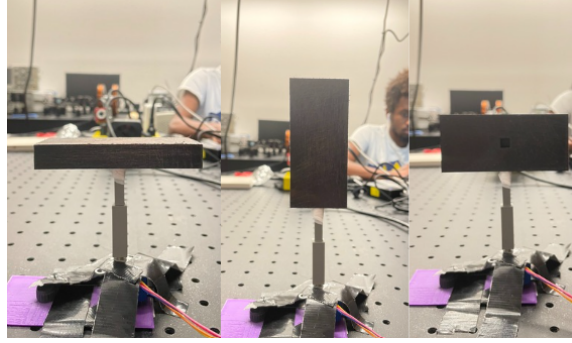


Figure 10: Rotation of the object about various principal axes, namely flat, upright, and standard, in that order.

5.2 Software Validation

5.2.1 Data Pre-processing and Automated Execution

The software incorporated a robust pre-processing pipeline that organized incoming data into a chronological sequence. Extraneous data points, such as those originating from the test stand or surrounding environment, were effectively pruned using spatial filtering techniques. The software automatically recognized new data inputs and executed the processing pipeline from start to finish, ensuring a smooth and efficient operation.

Pre-Processing



Post-Processing



Figure 11: Comparison of raw and processed point cloud data. Post-processing eliminates noise and outliers, leaving a clean and focused point cloud of the rectangular object.

5.2.2 Zero-Rotation and Rotation Identification

For objects that are not rotating, the system confirms the absence of significant changes in their position or orientation over time. For rotating objects, the system detects variations in orien-

tation between consecutive frames. By comparing sequential point clouds, the software reliably distinguishes between non-rotating bodies and those undergoing rotation.

```
Dataset: C:\Users\AMYES\OneDrive\Documents\GitHub\ComptuerEngineeringDesign\Camera_1rpm_Take1_10SecIntervals is moving
Dataset: C:\Users\AMYES\OneDrive\Documents\GitHub\ComptuerEngineeringDesign\NoMotion is not moving
```

Figure 12: Example of software output indicating whether the dataset represents a non-rotating object (“is not moving”) or a rotating object (“is moving”) based on changes in consecutive frames.

5.2.3 Shape and Dimension Recognition

For stationary objects fully within the LiDAR’s field of view, the system accurately identifies their silhouette and measures their primary dimensions (length, width, and depth). After data cleanup and downsampling, the resulting point cloud data yields reliable dimension measurements with minimal error.

Table 5: Rectangle dimension measurement error and likelihood

Dimension	Actual(cm)	Measured Mean (cm)	Std. Dev. (cm)	Mean Error (%)	Number of Measurements Within 5% Error	% Within 5% Error
Length	10.031	9.986	0.048	0.45%	4/4	100%
Width	4.524	4.551	0.033	0.57%	4/4	100%
Depth	1.523	1.469	0.079	3.54%	3/4	75%

5.2.4 Rotational Speed Estimation

When the object is rotating, the system determines the rotational speed by analyzing changes in its position and orientation across sequential frames. Using algorithms like Iterative Closest Point (ICP), the software can estimate rotation rates with a small margin of error.

Table 6: Measured Rotation Rates vs. Actual Values

Target RPM	Measured Mean (RPM)	Std. Dev. (RPM)	Mean Error (RPM)	Within ± 0.2 RPM (#/Total)	%Within ± 0.2 RPM
3.0	3.004	0.060	0.004	10/10	100%
1.0	1.009	0.066	0.009	10/10	100%

5.3 Troubleshooting and Mitigation Strategies

Throughout the testing process, several challenges were encountered and addressed:

5.3.1 Sensor Calibration and Alignment

Issue: Minor misalignments in the LiDAR sensor led to inconsistencies in the data.

Solution: A calibration procedure was established, involving the use of reference targets to adjust the sensor alignment before each test session. This ensured that the sensor's measurements were accurate and reliable.

5.3.2 Environmental Interference

Issue: Ambient lighting variations and mechanical vibrations introduced noise into the measurements.

Solution: The optics lab's blinds were used to control lighting conditions, minimizing the impact of external light sources. The LiDAR and camera were mounted on the optics table, which provided a stable platform to reduce vibrations.

5.3.3 Data Processing Efficiency

Issue: High-resolution data from the LiDAR sensor resulted in increased processing times, potentially affecting real-time performance.

Solution: Downsampling techniques were implemented to reduce the density of the point cloud data without significantly compromising accuracy. By selectively retaining critical data points, the processing load was reduced, and software performance was optimized.

5.3.4 Incomplete Sensor Integration Due to Equipment Delays

Issue: The delayed arrival of the camera lens prevented the integration of visual data into the system.

Solution: The team focused on enhancing the LiDAR data processing capabilities to compensate for the absence of camera data. Test objects were modified with surface treatments to improve LiDAR reflectivity and detail capture.

5.4 Validation Against Functional Requirements

The system was rigorously tested against the project's functional requirements, demonstrating compliance in the following areas:

- **Detection and Characterization:** Successfully detected and characterized both stationary and rotating objects, accurately determining their shapes and rotational parameters.
- **3D Modeling:** Generated 3D models of the test objects, facilitating comprehensive analysis and visualization.
- **Data Processing:** Implemented efficient algorithms for data pre-processing, noise reduction, and feature extraction, ensuring reliable performance.
- **System Stability:** Maintained stable operation through effective mechanical design and environmental controls, minimizing the impact of external factors.

5.5 Verification and Validation Results

The verification and validation efforts confirm that the CLiDA system meets its intended functional requirements and performs effectively under various test conditions. By addressing technical challenges through meticulous troubleshooting and strategic adaptations, the system demonstrates robustness and reliability.

More images and videos capturing the test setups, data visualizations, and system operation will be included to provide concrete evidence of the system’s performance and to illustrate the key findings discussed in this section before the final draft will be submitted.

6 Risk Assessment and Mitigation

6.1 Realized Risks and Their Impact

6.1.1 Errors in Software Interpretation of Simulated Rotational Data

This risk occurred frequently due to challenges in accurately interpreting rotational data from the simulations. The initial algorithms struggled with precise rotational analysis, leading to discrepancies in the calculated rotational speeds and orientations.

Impact and Mitigation To address this issue, we switched from using basic point cloud processing to utilizing the Open3D library, which offers advanced 3D data processing capabilities. The Open3D library provided more robust tools for handling rotational data and improved the accuracy of our interpretations. Despite this improvement, we continued refining our algorithms and conducted extensive testing with controlled rotation speeds to calibrate the system effectively.

6.2 Lessons Learned from Risk Management

The occurrence of these risks underscored several key lessons:

- **Importance of Early Equipment Acquisition:** Delays in receiving critical components like the camera can significantly impact the project timeline and scope. Future projects should prioritize early procurement and have contingency plans for equipment delays.
- **Flexibility in Tools and Methods:** Adapting to new software tools such as Open3D demonstrated the need for flexibility and willingness to explore alternative solutions when initial approaches encounter obstacles.

6.3 Alternative Actions

If additional time were available for this project, the following changes would be implemented:

- **Ensure Timely Acquisition of Components:** Implement a more proactive procurement schedule with buffer times for potential delays.
- **Develop Redundant Systems:** Incorporate backup sensors or data sources to mitigate the impact of missing components.

- **Enhance Simulation Environments:** Invest in creating more advanced simulation setups that can better mimic the conditions and characteristics of space debris.
- **Continuous Testing and Validation:** Regular testing at each development stage to identify and address issues promptly.

7 Project Planning

7.1 Organizational Structure

To ensure efficient project execution, clear roles and responsibilities were assigned to each team member. Table 7 outlines the team structure and individual responsibilities.

Table 7: Team Roles and Responsibilities

Role	Team Member	Responsibilities
Project Manager	Sahra Genc	Oversees overall project direction, coordination among team members, and ensures project milestones are met.
Hardware Lead Engineer	Jack Grimes	Manages hardware components, focusing on the stability and functionality of mechanical and electrical systems.
Testing Lead	Jerry Jack Jr.	Responsible for motor setup, creation of 3D objects for testing, and systematic data collection during experiments.
Software Lead Engineer	Elijah Smith	Leads software design, coding, and debugging processes, ensuring software components meet project requirements.
Algorithm Developer	Janga Bliss	Develops and tests algorithms for data processing, shape recognition, and visualization of LiDAR data.

7.2 Work Breakdown Structure (WBS)

Figure 13 illustrates the Work Breakdown Structure of the project, detailing the main tasks and sub-tasks required to deliver the CLiDA system. The WBS was developed to organize the team's efforts effectively and to ensure that all aspects of the project were addressed systematically.

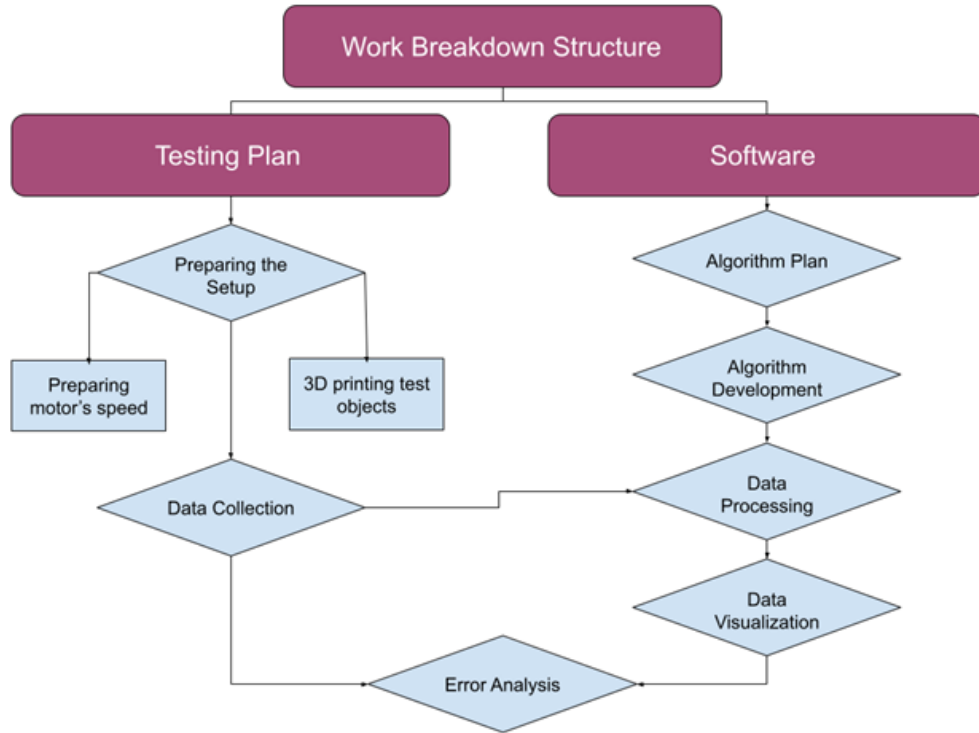


Figure 13: Work Breakdown Structure for the CLiDA Project

7.3 Work Plan (WP)

This Gantt chart illustrates the work plan for our junior design project, detailing the tasks and milestones necessary to achieve the Work Breakdown Structure (WBS) deliverables. The plan is divided into two main categories: the Testing Plan and Software Development. The Testing Plan includes purchasing components, preparing the test environment, setting up the motor, and designing a custom 3D object for validation. These tasks lead to the critical milestone of completing data collection by November 20, 2024. The Software Development phase involves algorithm planning, development, data processing, and visualization, with the final algorithm completed by November 18, 2024.

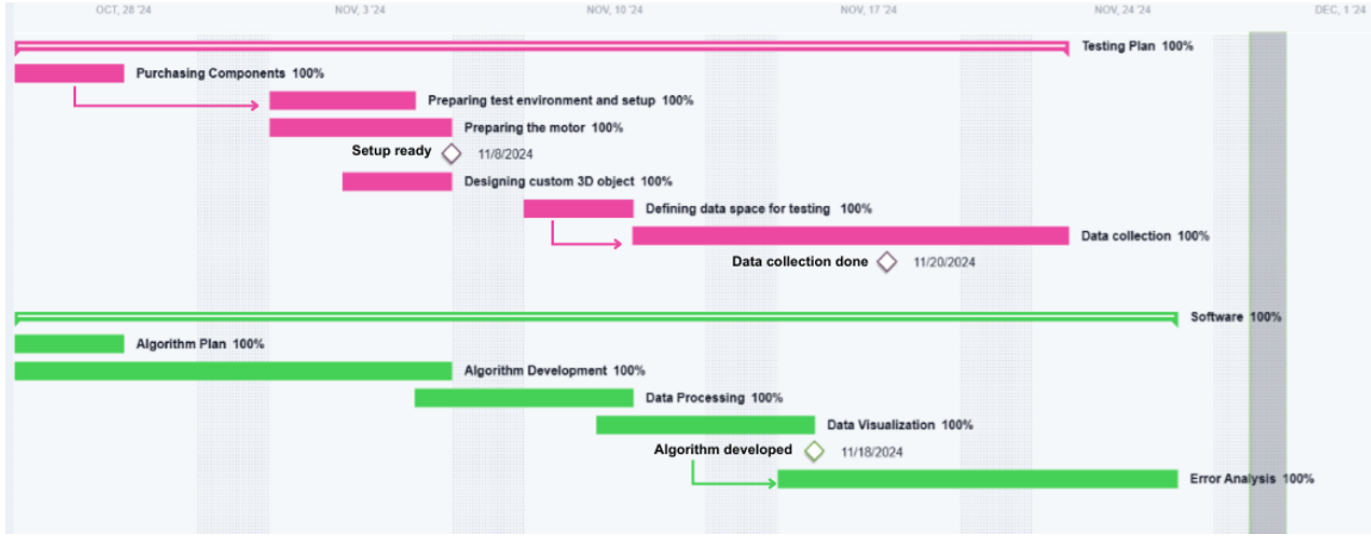


Figure 14: Gantt Chart for Project Timeline.

7.4 Cost Plan (CP)

A detailed cost plan was developed to manage the project's budget effectively. Table ?? presents the estimated expenses, including margins and uncertainties.

Initially, we commenced implementing our project utilizing the point cloud library. However, due to its outdated nature, we encountered challenges in applying it to our contemporary computers. Consequently, we transitioned to open3D, which exhibits a resemblance to PCL and operates on a more up-to-date system.

Item	Cost	Notes
Flash LiDAR Sensor	\$599.00	Purchased online; essential for 3D modeling.
Camera	\$380.99	High-resolution imaging; shipping included.
Miscellaneous Hardware	\$100.00	Camera lens, camera wire.
Final Total Cost	\$1079.99	Meets budget requirements.
Project Budget	\$2,000.00	

Figure 15: Estimated Project Budget and Expenses

7.5 Test Plan (TP)

The test plan was meticulously crafted to validate the CLiDA system’s performance against the project’s functional requirements and success criteria. It comprised structured studies designed to assess the system’s capabilities under various conditions. Access to specialized equipment and facilities, such as the optics lab and servo motor, was secured through coordination with laboratory management. The tests were scheduled to maximize resource utilization while ensuring thorough data collection. This test plan was drafted prior to the results being obtained and was an integral part of our planning phase for this project.

Explanation of Test Conditions:

- **Motion & Dimension Study:** Evaluated the system’s ability to detect and analyze a stationary object versus one in motion as well as the dimensions of the object. The rectangular object in upright orientation was observed at 0 RPM to establish a baseline and at 1 RPM to simulate motion.
- **Speed Study:** Investigated the impact of different rotational speeds on the system’s performance. The rectangular object was tested in both upright and flat orientations at 1 RPM and 3 RPM.

Table 8: Comprehensive Test Plan for CLiDA System Validation

Study Type	Test Condition	Object Shape	Orientation	Rotational Speed (RPM)
Motion Study	1 repetition	Rectangular	Upright	0
	10 repetitions	Rectangular	Upright	1
Speed Study	10 repetitions	Rectangular	Upright	1
	10 repetitions	Rectangular	Upright	3

7.5.1 Specialized Equipment and Facilities

The following specialized equipment and facilities were essential for conducting the tests:

- **Optics Laboratory:** Provided a controlled environment with minimal external interference.
- **LiDAR Sensor:** High-precision sensor for capturing detailed point cloud data.
- **Servo Motor:** Enabled precise control over the rotational speed of the test objects.
- **3D-Printed Objects:** Custom-designed rectangular and triangular shapes representing space debris.
- **Optics Table:** Offered a vibration-free platform for accurate measurements.

7.5.2 Data Collection and Analysis Procedures

For each test:

1. **Setup Verification:** Ensured all equipment was correctly set up and calibrated.
2. **Data Acquisition:** Collected LiDAR data with appropriate sampling rates.
3. **Repetition:** Conducted three trials per test condition to ensure data reliability.
4. **Data Storage:** Secured storage of all raw data for analysis.

Data analysis involved:

- Processing point cloud data using Open3D libraries.
- Applying noise reduction and filtering techniques.
- Comparing results across trials to identify standard deviation.
- Validating performance against expected outcomes.

7.5.3 Communication and Documentation

To maintain professionalism and ensure clarity:

- **Test Protocols:** Developed detailed procedures for consistency.
- **Roles Assignment:** Clearly defined team member responsibilities during tests.

Regular meetings facilitated:

- Discussion of test results and observations.
- Identification and resolution of issues.
- Project Documentation.

7.6 Risk Management Integration

Risk management considerations were integrated into the project planning process. Identified risks, such as equipment delays or data processing challenges, were addressed proactively through:

- **Contingency Planning:** Allocating additional time and resources for critical tasks.
- **Regular Monitoring:** Keeping track of potential risks and their impact on the schedule.
- **Communication:** Ensuring all team members were aware of risks and mitigation strategies.

7.7 Project Planning Summary

Effective project planning was crucial to the success of the CLiDA project. By defining clear roles, developing a comprehensive work plan, managing the budget diligently, and executing a thorough test plan, the team was able to meet project objectives and deliver a functioning system that simulates addressing the challenges of space debris detection and analysis.

8 Lessons Learned

The development of the CLiDA project provided invaluable insights and experiences for the team. This section highlights the key lessons learned throughout the project's lifecycle, including challenges faced, strategies employed, and recommendations for future projects.

8.1 Importance of Early Integration

One of the most significant lessons learned was the critical importance of initiating system integration early in the project timeline. By integrating hardware and software components at the earliest opportunity, the team was able to identify compatibility issues and technical challenges that may not have been apparent during the individual development phases.

- **Early Detection of Issues:** Early integration allowed for the timely detection of problems related to sensor communication, data synchronization, and mechanical alignment.
- **Iterative Improvement:** It provided the opportunity to iteratively improve the system through incremental testing and refinement.
- **Recommendation:** Future projects should allocate sufficient time for early and continuous integration efforts to mitigate risks and enhance overall system performance.

8.2 Effective Communication and Team Coordination

The project's success was heavily reliant on effective communication and coordination among team members.

- **Regular Meetings:** Holding regular team meetings ensured that all members were aligned on project goals, progress, and challenges.
- **Clear Role Definitions:** Clearly defined roles and responsibilities minimized overlaps and gaps in task execution.
- **Documentation:** Maintaining thorough documentation facilitated knowledge sharing and helped in onboarding any new team members or stakeholders.
- **Recommendation:** Establish strong communication channels from the outset, and utilize project management tools to keep the team organized and informed.

8.3 Flexibility and Adaptability

The team learned the value of being flexible and adaptable in the face of unforeseen challenges, such as equipment delays and technical difficulties.

- **Alternative Solutions:** When the camera lens did not arrive on time, the team adapted by enhancing LiDAR data processing and utilizing alternative materials to simulate solar panel characteristics.
- **Learning New Tools:** Adopting the Open3D library required the team to quickly learn and implement new software tools, which ultimately improved data processing capabilities.

- **Recommendation:** Embrace change and be prepared to pivot strategies as necessary. Building a culture of adaptability can turn potential setbacks into opportunities for innovation.

8.4 Thorough Testing and Validation

Comprehensive testing and validation were essential in ensuring the system met its functional requirements.

- **Structured Test Plan:** Developing a detailed test plan helped in systematically evaluating system performance under various conditions.
- **Data Quality:** Emphasis on data accuracy and validation against known measurements enhanced the reliability of the results.
- **Recommendation:** Allocate ample time for testing phases and ensure that test plans are well-structured, with clear objectives and expected outcomes.

8.5 Challenges of Sensor Integration

Integrating multiple sensors presented technical challenges that required careful consideration.

- **Synchronization Issues:** Aligning data streams from the LiDAR sensor and the camera proved more complex than anticipated.
- **Calibration Needs:** Regular calibration was necessary to maintain sensor accuracy, highlighting the need for calibration protocols.
- **Recommendation:** Future projects should plan for the complexities of sensor integration, including synchronization mechanisms and calibration routines.

8.6 Realistic Simulation Environments

Creating realistic simulation environments was more challenging than initially expected.

- **Material Limitations:** Simulating the reflective properties of solar panels without the camera limited the realism of tests.
- **Environmental Factors:** Replicating space conditions, such as low-light environments, required solutions like using optics lab blinds.
- **Recommendation:** Invest time in developing accurate simulation environments and consider all environmental variables that may impact system performance.

8.7 Time Management and Scheduling

Effective time management was crucial, especially when faced with unexpected delays.

- **Equipment Delays:** Delays in receiving critical components like the camera lens impacted the project schedule.

- **Task Prioritization:** The team had to reprioritize tasks to keep the project moving forward despite setbacks.
- **Recommendation:** Build buffer times into the project schedule to account for potential delays, and regularly reassess priorities to adapt to changing circumstances.

8.8 Advice for Future Teams

Based on our experiences, we offer the following advice to new juniors embarking on similar projects:

- **Start Early:** Begin integration and testing as soon as possible to uncover unforeseen challenges.
- **Maintain Open Communication:** Foster a collaborative environment where team members feel comfortable sharing ideas and concerns.
- **Document Everything:** Keep detailed records of designs, decisions, and changes to aid in troubleshooting and future reference on Github.
- **Be Proactive with Procurement:** Order critical components early and have contingency plans in place.
- **Embrace Learning:** Be prepared to learn new tools and technologies as the project evolves.
- **Stay Flexible:** Be willing to adjust plans and approaches in response to new information or challenges.

8.9 Summary of Lessons Learned

The CLiDA project was a valuable learning experience that enhanced our technical skills and project management abilities. By reflecting on the lessons learned, we aim to apply this knowledge to future endeavors and contribute to the continuous improvement of engineering practices.

9 Individual Report Contributions

9.1 Jerry Jack

Developed and implemented basic Arduino-based motor control to ensure precise test rotations. Assisted in preparing 3D-printed test objects, applying a sanded finish to reduce LiDAR scattering, and supported initial data collection efforts.

9.2 Janga Bliss

Significantly contributed to the software side of the project, developing and debugging the initial LiDAR processing code to enable RPM detection, axis identification, dimension measurement, data pruning, and file format conversions. He also 3D-printed silhouettes and mounts for stable test configurations, ensuring consistent sensor alignment. In addition to these technical responsibilities, authored sections 4, parts of 5, 6, parts of 7, and 8 of the report, shaping a substantial portion of the project's documentation.

9.3 Elijah Smith

Established and maintained the coding environment while managing version control via GitHub, optimizing the development workflow and ensuring seamless traceability. Addressed challenges in data collection and 3D printing, created point cloud visualizations to extract deeper insights, and integrated camera and LiDAR data for a comprehensive understanding of the debris. These technical contributions were instrumental in refining and validating the system's data analysis capabilities. Due to the late arrival of the camera, dedicated significant time to troubleshooting and programming it to detect RPM during our final week ensuring the project delivered meaningful results.

9.4 Sahra Genc

Coordinated the project's day-to-day operations, ensuring alignment with milestones and seamless communication among team members and advisors. Contributed to the baseline design, developed testing setups, assisted with algorithm development, and designed 3D-printed objects using Solid-Works. Defined data spaces and performed critical data interpretation, guiding the project toward its objectives. Written contributions include authoring sections 1, 2, 3, parts of 5, and parts of 7, providing structure and coherence to the final report. Documented all of our results on the report and documented the results on the presentation as well as the processing unit slides.

9.5 Jack Grimes

Assisted in establishing the baseline design, contributed to data collection, helped design 3D-printed objects, and aided in defining data parameters. Played a supporting role in setting up and configuring the hardware testing environment. Designed the structure and format of the presentation, ensuring it flowed well while also conveying the information needed. Assisted in establishing the baseline design, contributed to data collection, helped design 3D-printed objects, and aided in defining data parameters. Played a supporting role in setting up and configuring the hardware testing environment.

10 Appendices

10.1 Key Design Alternatives

10.1.1 Design Option 1: Flash lidar

This design uses a flash lidar system to capture a point cloud map of the solar panel. The point cloud data allows us to build a 3D model of the solar panel, which we can run through an algorithm to determine the rotational speed and axis. The 3D model will also give us the object's size and shape. The flash lidar allows us to model the object much easier and determine its rotational state with a higher accuracy. The Point Cloud Library algorithm can be integrated to analyze the data and to identify solar panels by shape and pattern.

Table 9: Pros and cons tables for design option 1

Pros	Cons
More effective in low light conditions	More expensive
Access to more pre-built algorithms and simulations to work with point clouds	Trouble detecting transparent or dark materials such as glass.
Higher quality and more accurate outputs with 3D models of the object.	Noisier results when testing on earth (extra light clutter compared to space)

10.1.2 Design Option 2: Camera with simple lidar

This second design implements a simple lidar system acting as a range finder. The camera captures series of images of the solar panel while the lidar measures its distance. The light source is essential to improve the visibility in a low-light condition to ensure that the camera can capture a clear image of the solar panel. The lidar's role in this design more basic compared to the flash lidar system. It provides distance measurements but does not generate detailed point cloud data. The collected data from the camera and the lidar can be used to estimate the solar panel's size and position, but with lower precision in terms of rotational state or detailed shape modeling.

Table 10: Pros and cons tables for design option 2

Pros	Cons
Low Cost	Increased complexity with multiple components
Fewer data points/lower data processing	Lower shape accuracy

10.2 Trade Study Process and Results

Table 11: Consideration and Weights for Design Options

Metric	Weight	Driving Requirements	Description
Structural Integrity	0.1	FR1: DR1-3	Structural integrity is important to avoid vibrational errors in data.
Operation	0.3	FR2:DR1-3	Operation of the design must work as expected to collect the data needed for the simulation.
Shape Recognition	0.2	FR3:DR1-3	Shape recognition is must be functioning on the design to identify the target object.
Simulation	0.3	FR4:DR1-3	The simulation must be able to run so to determine the rotation of the object.
Interface	0.1	FR5:DR1-3	The interface must be straightforward to use to provide a simple data collection.

Table 12: Metric Ranking

Metric	1	2	3	4	5
Structural Integrity	The setup is not mounted and not stable	The setup is loosely mounted, so not stable.	The setup is mounted and stable with high vibrational errors.	The setup is mounted and stable with medium vibrational errors.	The setup is mounted on an optics table and there is minimal vibrational error in the data.
Operation	The system is not capable of collecting data.	The system is capable of collecting data. Is not capable of storing data.	The system can collect data but inconsistent when storing data.	The system can collect and store data of the object when the earth is not in view.	The system can collect and store data of the object regardless of background.
Shape Recognition	The system is not capable of recognizing the shape of the object it sees.	The system can recognize an object, but the shape is incorrect.	The system can recognize the shape of the target object but not its dimensions, color, or texture.	The system is capable of recognizing the shape and the color of the object but not its dimensions.	The system is capable of recognizing the shape of the target object and its dimensions, color and texture of the material.
Simulation	The system does not determine the rotation of the object	A 2D contour map is created from the received lidar points.	A 2D model is created of the object showing its position to the lidar.	The rotational state of the object is determined using an algorithm	A 3D model of the object is created, and tells us the object's rotational state
Interface	There is no user interface.	The interface is clunky.	The interface is functional.	The interface is user-friendly to a programmer.	The interface is user-friendly with no coding knowledge.

Table 13: Trade Study Evaluation

Metrics	Structural Integrity	Operation	Shape Recognition	Simulation	Interface	Total
Weight	0.1	0.3	0.2	0.3	0.1	1
Camera with standard LiDAR and light source	5	4	4	4	5	4.2
Flash LiDAR with a standard camera	4	5	5	5	5	4.9

References

- [1] S. Ivanov, B. Konstantinov, S. Tzokov, T. Ivanov, T. Kanev, V. Zlateva, M. Avramov, B. Ivanov, N. Neshev, V. Vassilev, and O. Ognyanov, "Space Debris Identification, Classification and Aggregation with Optimized Satellite Swarms," Conference on Space Debris Management, 2016.
- [2] D. Mehrholz, L. Leushacke, W. Flury, R. Jehn, H. Klinkrad, and M. Landgraf, "Detecting, Tracking and Imaging Space Debris," ESA Bulletin, no. 109, pp. 128-134, Feb. 2002.
- [3] M. V. H. Khan, "Space Debris: Overview and Mitigation Strategies," 2024.
- [4] A. Murtaza, S. J. H. Pirzada, T. Xu, and L. Jianwei, "Orbital Debris Threat for Space Sustainability and Way Forward (Review Article)," IEEE Access, vol. 8, pp. 61000–61019, 2020, doi: <https://doi.org/10.1109/access.2020.2979505>.
- [5] B. Dadhich, R. Guha, S. Shekhar, and V. Malhotra, "Autonomous Space Debris Capturing System for Recycling," 2021 IEEE Aerospace Conference (50100), Mar. 2021, doi: <https://doi.org/10.1109/aero50100.2021.9438495>.
- [6] M. Shan, J. Guo, and E. Gill, "Deployment dynamics of tethered-net for space debris removal," Acta Astronautica, vol. 132, pp. 293–302, Mar. 2017, doi: <https://doi.org/10.1016/j.actaastro.2017.01.001>.
- [7] S. M. Group, "Spacecraft to Remove Orbital Debris," www.techbriefs.com, Dec. 01, 2022. <https://www.techbriefs.com/component/content/article/47123-msc-tops-90>
- [8] European Space Agency, "ClearSpace-1," https://www.esa.int/Space_Safety/ClearSpace-1
- [9] "RemoveDEBRIS | Airbus," <https://www.airbus.com/en/products-services/space/in-space-infrastructure/removedebris>